

Let's Practice

IF Statement



Example 1

```
always @ (*)
begin
  if( done_flag == 2'b00)
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
  else if ( done_flag == 2'b01)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
    end
  else
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
end
```

Example 2

```
always @ (*)
begin
  if(E)
    begin
      B = 1'b0 ;
      C = 1'b0 ;
    end
  else if (F)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
      C = 1'b1 ;
    end
  else if (G)
    begin
      A = 1'b0 ;
    end
  else
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
end
```

Example 3

```
always @ (*)
begin
  if( done_flag == 2'b00)
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
  else if ( done_flag == 2'b01)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
    end
  else if ( done_flag == 2'b11)
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
end
end
```

Example 4

```
always @ (*)
begin
  if( done_flag == 2'b00)
    begin
      A = 1'b1 ;
      B = 1'b0 ;
      C = 1'b0 ;
    end
  else if ( done_flag == 2'b01)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
      C = 1'b0 ;
    end
  else
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
end
```

Let's Practice

Case Statement



Example 1

```
always @ (*)
case (sel)
  2'b00: begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
  end
  2'b01: begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
  end
  2'b10: begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
  end
endcase
```

Example 2

```
always @ (*)
case (sel)
  2'b00: begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
  end
  2'b01: begin
    B = 1'b1 ;
  end
  2'b10: begin
    A = 1'b1 ;
  end
  2'b11: begin
    A = 1'b1 ;
    B = 1'b0 ;
  end
endcase
```


Example 3

```
always @ (*)
begin
    A = 1'b1 ;
    B = 1'b0 ;
case (sel)
2'b00: begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
    end
2'b01: begin
    B = 1'b1 ;
    end
2'b10: begin
    A = 1'b1 ;
    end
default: begin
    A = 1'b1 ;
    B = 1'b0 ;
    end
endcase
end
```

Example 4

```
always @ (*)
begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
case (sel)
    2'b00: begin
        A = 1'b1 ;
        end
    2'b01: begin
        B = 1'b1 ;
        end
    2'b10: begin
        C = 1'b1 ;
        end
    default: begin
        A = 1'b1 ;
        B = 1'b0 ;
        end
endcase
end
```